Two-Proportion Test

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Introduction

The two-proportion test compares success rates in two independent groups. Typical applications: treatment vs. control event rates, exposed vs. unexposed incidence, screen-positive rates across populations. The test comes in two algebraically equivalent forms – a z-test on the difference and a 2x2 chi-squared – and is typically reported together with a risk difference, relative risk, or odds ratio.

Prerequisites

One-proportion test, chi-squared distribution.

Theory

For $X_1 \sim \text{Binomial}(n_1, p_1)$ and $X_2 \sim \text{Binomial}(n_2, p_2)$ independent,

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_1 : p_1 \neq p_2.$$

Pooled proportion under H_0: $\hat{p} = (x_1 + x_2)/(n_1 + n_2)$.

Z-test:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}.$$

Z^2 equals the chi-squared statistic on the 2x2 table. In R, `prop.test()` performs this test (using chi-squared form).

Effect-size summaries:

- Risk difference (RD): $\hat{p}_1 - \hat{p}_2$.
- Relative risk (RR): $\hat{p}_1/\hat{p}_2$.
- Odds ratio (OR): $\hat{p}_1(1-\hat{p}_2)/[\hat{p}_2(1-\hat{p}_1)]$.

Each has a standard-error formula for constructing Wald CIs; score-based and exact alternatives exist.

Assumptions

- Independent Bernoulli trials within and between groups.
- Expected counts ≥ 5 in every cell (else use Fisher's exact test).

R Implementation

```r
library(epitools)

x1 <- 48; n1 <- 200        # treatment
x2 <- 32; n2 <- 200        # control
```

```r
prop.test(c(x1, x2), c(n1, n2), correct = FALSE)

# Risk difference with 95% CI via prop.test
prop.test(c(x1, x2), c(n1, n2), correct = FALSE)$conf.int

# Relative risk and odds ratio
riskratio(c(n2 - x2, n1 - x1, x2, x1), rev = "both")$measure
oddsratio(c(n2 - x2, n1 - x1, x2, x1), rev = "both")$measure

# Fisher's exact test (small-cell alternative)
fisher.test(matrix(c(x1, n1 - x1, x2, n2 - x2), nrow = 2, byrow = TRUE))
```

Output & Results

```
     2-sample test for equality of proportions without continuity correction
X-squared = 3.66, df = 1, p-value = 0.0557
95% CI for (p1 - p2): (-0.0014, 0.1614)
sample estimates: prop 1 = 0.24, prop 2 = 0.16

           estimate lower    upper
Exposed        1.50 0.994     2.265
              (relative risk 1.50)
```

Treatment success rate 24 %, control 16 %. Risk difference 8 % (95 % CI -0.1 % to 16.1 %); p = 0.056 (borderline). Relative risk 1.50 (95 % CI 0.99 to 2.27).

Interpretation

Report counts, proportions, the effect-size estimate with its CI, and the test statistic. "In the treatment arm, 48/200 (24.0 %) experienced the event, compared to 32/200 (16.0 %) in control (risk difference 8.0 %, 95 % CI -0.1 % to 16.1 %; $\chi^2(1) = 3.66$, p = 0.056)."

Practical Tips

- Report at least one of RD, RR, or OR as an effect size; significance alone is unhelpful.
- Use Fisher's exact test when expected counts are below 5.
- For matched/paired binary data, use McNemar's test, not the two-proportion test.
- The chi-squared test and the two-proportion z-test give identical two-sided p-values when `correct = FALSE`.
- Be aware that RR and OR differ in magnitude when outcomes are common ($p > 0.10$); choose the scale that fits the clinical question.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem

- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests